

The Missing Medium: A Transactional Residual Stream for Transformer Training

Hyphae Research Foundation

Version 1.1 — 5 September 2026 (v1.0: 4 September; reframed after four external reviews, see [paper/notes/REVIEWS-2026-09.md](#)).

Code: Apache-2.0. This document, its figures and the results it cites: CC BY-SA 4.0.

Every number traces to a file in **results/** or a record key in a HYPHAE ledger (§12).

Abstract

Modern training treats the transformer residual stream as an anonymous accumulator: any kernel may execute $h += \delta$, and no record of the write survives the step. Training observes the *consequences* of internal writes — loss curves, evaluations, activation probes — never the writes themselves. We introduce HYTORCH, a training loop in which the residual stream is a *typed, transactional channel*: blocks propose, a pinned policy admits, and every candidate write leaves exactly one 16-byte record — COMMIT, OVERFLOW (lost a slot contention) or ABORT (rule rejection) — so that what the model *attempted* to write, and why it was refused, is data. Every optimizer step is gated by a durable receipt from an embedded transactional store (HYPHAE); a CPU-only verifier replays audited microbatches bit-for-bit against a pinned software reference; the same reference binary admits NVIDIA (sm_89/sm_90), AMD (gfx950) and, through a two-phase policy, AWS Trainium2 kernels on 200,000 adversarial cases with bit-identical verdicts and residuals. Fault injection over real training spills is detected 36/36 with the correct class.

The instrument produced two results. Our preregistered protocol reversed our own headline: a single-LR –44.9% perplexity win, replicated across vendors, became a 2.9× loss to the properly tuned dense twin under the preregistered LR sweep, and we publish the reversal. And at nanochat-d20 scale the ledger recorded a *silent channel collapse*: commit rate 0.066% → 0 by step 200, the write codebook frozen bit-exactly, while validation loss improved on schedule through the architecture’s declared bypass paths. Neither loss, evaluation nor activation probes distinguish an expensive pathway from a dead one; the write records do, mechanically, during training. We are explicit about what this is and is not. HYTORCH is an instrumentation and verification framework, not an efficient training architecture: the channel configuration we tested

(at most 160 scalar degrees of freedom per token per layer) does not carry a language model — the cataloged d20 model is at chance on knowledge tasks — and costs 12–29× the twin’s wall clock; its headline capacity number comes from a run whose STE gradient we later found to be mis-scaled, and the corrected rerun does not yet exist. What the evidence supports is narrower and, we argue, more useful: a training process whose state transitions through the typed channel are witnessed, receipted and replayable.

1 Introduction

A transformer’s residual stream is its only inter-layer medium: every head and every MLP communicates by adding into it. Yet in every production training stack this medium is an *anonymous accumulator*. Writes carry no identity, no admission decision, no durable trace; the optimizer steps whether or not anything about those writes was ever inspected. All observability is reconstructed after the fact from the outside (loss curves, evals, activation probes) — none of it witnesses the *acts*.

HYTORCH makes the acts the unit of record. The residual stream becomes a typed, transactional channel: blocks *propose* deltas; a pinned policy object packs proposals against a learned write codebook, allocates winners per slot, and *applies* only committed bindings. Every candidate — including every rejected one — becomes a 16-byte record in a durable ledger, and `optimizer.step()` is legal only after the step’s records have a commit receipt. Figure 1 shows the seam. To our knowledge, prior systems do not combine residual-write journaling, durable per-step receipts and bit-exact replay at this granularity (§10); we do not claim more priority than that.

Three words recur below and are used in their narrow senses. *Observability*: we know which writes occurred and which were refused. *Auditability*: we can show they occurred and were not altered (tamper-

evident, not unforgeable). *Replayability*: a CPU can reproduce them bit for bit. We make no claim of *interpretability* (what a write means) beyond one small probe, and no claim of *causality* (that a write causes a behaviour) beyond one journalized intervention; both are marked where they appear.

The contributions are:

- **C1 (transactional residual stream)**: a training loop where residual writes are typed transactions with verdicts, the negative space (OVERFLOW, ABORT) is recorded as data, receipts gate the optimizer, and a CPU replays audited steps bit-exactly (§2–4).
- **C2 (deterministic cross-silicon semantics)**: one reference binary admits NVIDIA and AMD kernels to *bit-identical* verdicts and residuals on 200k adversarial cases; the port to AWS Trainium2 moved the verification boundary without weakening it, and the channel trains there under DDP with every step receipted (§3, §8).
- **C3 (mechanical observability, including of ourselves)**: 36/36 injected faults caught with the correct class, and a preregistered evaluation whose sweep reversed our own replicated headline (§5).
- **C4 (the finding)**: silent channel collapse — a load-bearing internal pathway can die completely while training telemetry reads healthy, because the host architecture carries the trunk through declared bypass paths. Write-level records distinguish this from a merely expensive pathway during training; observational activation evidence alone cannot (§6).
- **C5 (what it costs)**: with the channel verifiably alive at d20, the configuration tested does not carry a language model and costs $12\times$ (exact kernels) to $29\times$ (two-phase) the twin’s wall clock; we report this as the measured price of the instrument as configured, with the caveats it carries (§7).

In one line: make the writes observable (C1), make them deterministic (C2), make the experiment adversarial to our own claims (C3), show what becomes visible once the writes are witnessed (C4), and say what it costs (C5).

2 Design: five laws

Law 0 (single writer, declared exceptions). Within a block, the catalog write is the only mutation of h : attention and MLP outputs are *proposals* δ , not writes. Host architectures may carry residual mutations *outside* the block (nanochat’s per-layer

$x \leftarrow \lambda_i x + \mu_i x_0$ scalars and smear/backout gates). These are not silenced and not pretended away: they are declared in the manifest and journalized every step as a BYPASS fact (the scalar values), so an auditor sees exactly which writes went through the typed channel and which did not. §6 shows why this matters.

Law 1 (typed writes). A write is a binding ($pos, slot, feature, mag$): magnitude $\times$ unit-normalized codebook row, applied to one slot of one token. k candidates per token; one winner per slot.

Law 2 (verdicts are facts). Every candidate leaves exactly one fact: COMMIT, OVERFLOW (lost a slot collision), or ABORT (nonfinite, magnitude cap, or policy denial). The negative space is journalized, not dropped: $\#C + \#O + \#A = k$, always.

Law 3 (receipts gate the step). All ranks spool their frames; rank 0 waits for the ledger’s commit receipt and broadcasts its head; a missing receipt kills all ranks. There are no ledger-less steps — we verified this the hard way (§6, incidents).

Law 4 (replayability). The bf16 bit-contract (promote, fp32 sequential no-FMA reductions, RNE downcast, canonical NaN) is pinned in a reference object whose SHA-256 is cited by every run. A CPU replay of any audited microbatch reproduces residuals bit-for-bit.

3 The pinned reference and per-silicon gates

The policy (pack $\rightarrow$ allocate $\rightarrow$ apply) exists once, as a Rust reference compiled to `libapply_ref.so`; CUDA (sm_89/sm_90) and HIP (gfx950) kernels are *admitted* per silicon by a differential harness: 200,000 adversarial cases in total, drawn round-robin from seven generator families (NaN/Inf floods, denormals, cap boundaries, tie storms, ...) under a fixed seed — i.e. $\sim 28k$ per family, reproducible rather than fresh per run — must produce byte-identical verdict records and residuals. A single canonicalization rule (NaN $\rightarrow$ +qNaN before ordering) was required to make x86 and device NaN semantics agree — found by the gate, not by inspection. The kernels use a packed ($key, feature$) comparator so parallel argmax equals the reference’s sequential tiebreak exactly.

Fault injection (Fig. 2): 36 mutations across six classes (bit-flipped magnitudes, dropped/injected commits, reordered application, wrong codebook row, truncated wire) against real training spills; 36/36 detected, each with the correct failure class (head mismatch, apply error, or chain break).

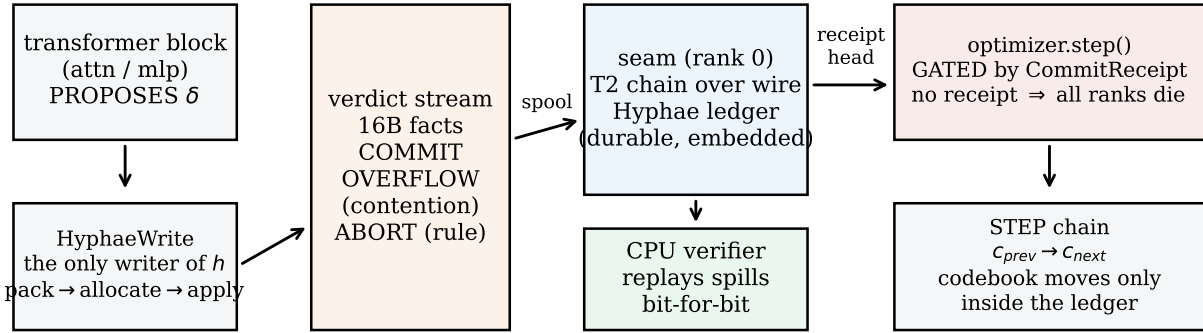


Figure 1: The HYTORCH seam. Blocks propose; HyphaeWrite is the only writer of the residual. The verdict stream (16-byte facts: commits, contentions, rule rejections) is spooled to a rank-0 seam that chains it (T2) into an embedded HYPHAE ledger. The optimizer is gated by the commit receipt; the codebook may move only inside the ledger ($C_{prev} \rightarrow C_{next}$ chain). A CPU verifier replays audited microbatches bit-for-bit.

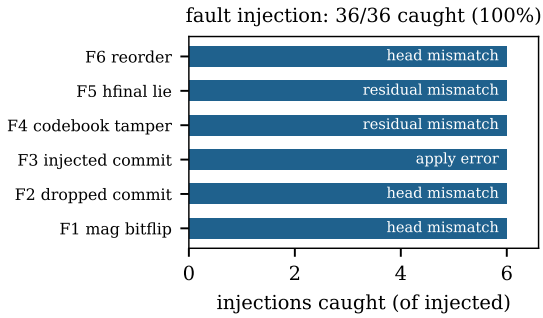


Figure 2: Fault injection over real training spills: 36/36 caught, correct class each. `results/inject/inject-summary.json`.

4 Authority $\neq$ durability: the seam

The trainer holds *authority* (it computes); the ledger holds *durability* (it remembers). The seam is a rank-0 process consuming per-layer wire frames, walking a hash chain (T2) over every fact, and committing {layer records, receipt, step chain} transactionally into HYPHAE, an embedded store. Two costs must be kept apart. The *seam* overhead — spooling, chaining, committing — is 9.5% of wall clock at 20k steps on one GPU and $\sim 5\%$ amortized at nanochat-d20 $\times 8$ ranks (335M facts/step; the T2 chain and counts are durable for *every* step, raw wire sampled at a declared cadence). The *channel* cost — running pack/allocate/apply for 40 write units over a 32,768-feature codebook in the forward and its mirror in the backward — is far larger: at d20

on 8 $\times$ MI355X the catalog arm ran at **7.42 s/step vs. 0.62 s/step** for the dense twin (12 $\times$ wall clock; `results/models/d20-3.4/NOTES.md`), with the exact device kernels executing the full N_f scan. Two-phase selection (§8) removes the exact scan from the device but pays a host round trip per write unit instead: 29 $\times$ at d10 on Trainium2. Readers should take 12 $\times$ (exact kernels) to 29 $\times$ (two-phase), not 9.5%, as the current end-to-end price of the typed channel.

Records scale: the d20 runs produce 3.4×10^8 verdict records per optimizer step (1.7×10^{12} per full run). What is durable for *every* step is the T2 chain head, the per-layer counts and the receipt; the raw 16-byte records themselves (~ 55 TB per run at full cadence) are retained at a declared cadence (every 100 steps). “Receipted on 100% of steps” therefore means every step’s writes are chained and countable, not that the full transcript is stored.

What a receipt guarantees. The receipt is the head of an unkeyed SHA-256 chain over every fact of the step, stored transactionally. It is *tamper-evident*: any post-hoc modification, drop, reorder or injection of a fact is detected (§3, 36/36). It is not *unforgeable*: an adversary who controls the seam process could fabricate an internally consistent chain from scratch. Unforgeability requires keying or external anchoring of heads (a straightforward extension we have not built); every claim in this paper is an integrity claim, not a non-repudiation claim.

5 Preregistration catches our own headline

Phase 1’s preregistered gate (PPL gap $\leq 10\%$ vs a *topology-matched* dense twin — same depth, width, data and schedule; the catalog arm carries $\sim 3.5\%$ more theoretical FLOPs and an order of magnitude more wall clock, so “FLOPs-matched” would overstate the parity — wikitext-103) **failed** honestly (+11.17%); a spec amendment (removing a ReZero-style gate that stabilized the dense twin) produced a **pass**: catalog 167.8 vs twin 304.7 (−44.9%), replicated on AMD (−43.2%), 930/930 audited microbatches green across three seeds.

The preregistration also demanded an LR sweep before any claim. Figure 3 shows what it found: the dense twin at its own best LR beats the catalog 2.9 $\times$; the −44.9% was an artifact of comparing both arms at the one LR where the twin destabilizes. What survives is real but different: the catalog is 6 $\times$ more LR-robust and 7 $\times$ more seed-stable (Fig. 4) — a stabilizer, not a free win. We publish the reversal; C1/C2 never depended on winning perplexity.

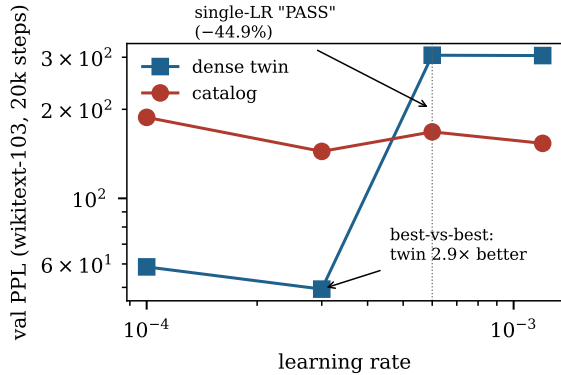


Figure 3: The preregistered LR sweep reverses the single-LR headline. [results/train/NVIDIA-H200/lrsweep-*/NOTES.md](#).

6 The finding: silent channel collapse

Scaling to nanochat-d20 (1.9B-token compute-optimal horizon, $8\times\text{MI355X}$, $d_{\text{model}}=1280$, 40 write units, $k=8$, magnitude cap 64), the ledger recorded the following (Fig. 5): at step 100 the commit rate was 0.066% with surviving magnitudes pinned to the cap ($p50 = 58$ of 64); by step 200 it was *exactly zero* and stayed zero. Proposals grew to 10^5 – 3.7×10^6 (up to 58,000 $\times$ the cap). With zero commits the straight-through gradient path dies; the codebook’s STEP chain shows $c_{\text{prev}} = c_{\text{next}}$ — frozen bit-exactly.

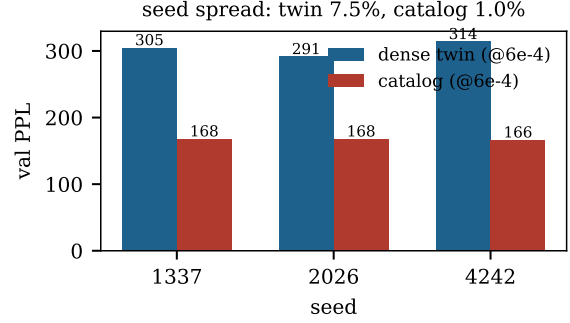


Figure 4: What survived the sweep: the catalog’s seed variance is 7 $\times$ smaller than the dense twin’s (at the twin-unstable LR).

Meanwhile validation bits-per-byte improved on schedule (3.17 $\rightarrow$ 1.32 by step 750): nanochat’s bypass parameters (x_0 /residual scalars, value embeddings) let the model learn *around* its dead channel.

Mechanism: raw attention/MLP outputs at $d=1280$ already sit near the cap at init; LR warmup pushes the distribution through it; overflow/abort $\Rightarrow$ no gradient $\Rightarrow$ no correction — a one-way feedback loop into permanent death. The fix is a per-slot proposal norm clip at cap/2 before the pack (direction preserved); with it the channel holds a 20–40% commit rate at the same scale (Fig. 5c), with the collapse telemetry now printed in the trainer log every 50 steps.

This is not a new phenomenon: it is codebook collapse (VQ-VAE) and router collapse (mixture-of-experts) wearing a training-time mask, and a smaller instance had already occurred in phase 1 at cap 8 (“COMMIT=2, OVERFLOW=118,986, loss kept dropping”), then “fixed” by raising the cap the toy inherited. What *is* new is that the ledger exposes it mechanically and that our phase-1 manifests had preregistered a codebook-usage-entropy kill threshold that no code implemented — a gap between promise and guard that this incident closed: the threshold (8 bits usage entropy, 2% commit rate, sustained 50 steps) now executes inside the seam and kills all ranks.

Three consequences follow. First, our earlier d20 “capacity cost” measurements were measurements of a bypass model, and we reinterpret them as such. Second, the C1/C3 thesis is demonstrated on this stack: “expensive channel” and “dead channel with bypass” are indistinguishable from outside and mechanically distinct in the records. Third, a consequence for mechanistic interpretability, stated at the strength the evidence supports: a claim of the form “this observed mechanism explains the behaviour” cannot be established from observational activation evidence alone, because the model may be using another path — here it was. Interventional methods (acti-

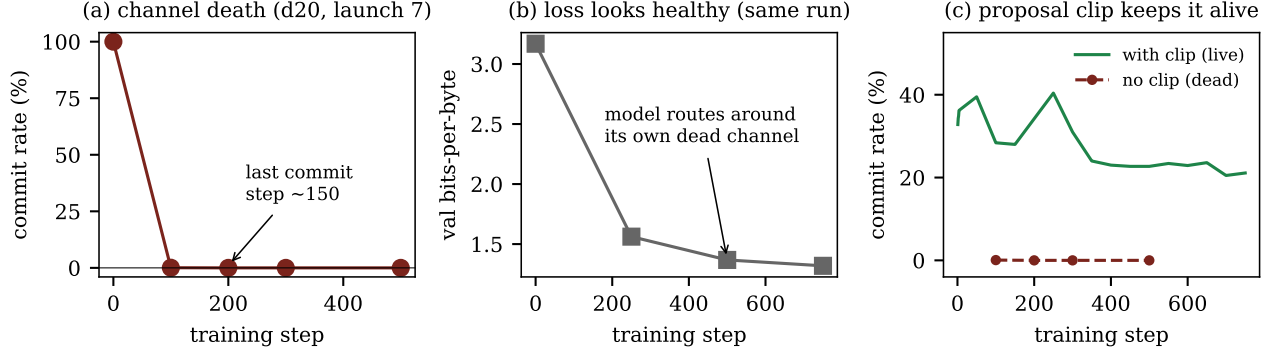


Figure 5: Silent channel collapse at d20. (a) Commit rate measured from retained wire (335M facts/step): dead by step 150–200. (b) The same run’s validation loss improves normally — the model routes around the dead channel. (c) With the proposal clip the channel stays alive at the same scale. `results/nanochat/PHASE5-CHANNEL-COLLAPSE.md`.

vation patching, ablation, causal tracing) can establish it; a write ledger establishes it for the writes it witnesses, which in HYTORCH are those of the typed channel and the declared bypass scalars, not every path. Our sibling project independently adopted a grad-norm “dead channel guard” after this incident; it catches the total-freeze case but not partial bypass — only per-write verdicts see that.

Two things this finding is not. It is not a discovery of collapse: VQ codebooks and MoE routers collapse, and the literature has remedies (§10). What is new is the ability to tell, during training and from write-level evidence, a dead internal pathway from an expensive functioning one. Nor is it yet a law of the residual stream. The host is nanochat, whose per-layer $x \leftarrow \lambda_i x + \mu_i x_0$ scalars are a trunk sufficient to carry the loss; in a pre-LN architecture without such scalars a dead channel might break the loss instead of hiding. How much of the silence is Law 0’s declared bypass and how much is the VQ bottleneck is open; the guard is the same in either case. Phrases such as “the model learned to route around its channel” are our interpretation of the records (commits at zero, bypass scalars journalized and moving, loss falling); the records establish the correlation, not the mechanism.

7 What it costs

A caveat before the number. The d20 pretraining reported here ran with a gradient bug we found later, while chasing an SFT-stage NaN: the proposal clip was applied to the *detached* input inside the write Function, so the STE mirror returned gradients for the clipped proposal that flowed into the unclipped block output, over-weighted by the clip ratio (up to 400× at d20) and compounding to *inf* within 13 layers on the fine-tuning distribution. The fix (clip in-graph, so the policy judges exactly what autograd sees) is in

the code and in the Trainium2 run of §8; the d20 re-run with the corrected mirror does not yet exist. The records and receipts are unaffected — the ledger stores what was written, not what gradient was returned — but Table 1 is the result of *that* run and must be read as such: because this run predates the corrected STE mirror, CORE 0.015 should not be interpreted as a clean estimate of the configuration’s capacity ceiling — it is the observed result of the current live-channel run, confounded by the now-identified bug, and the rerun is the definitive measurement. We keep it because it is the only full-horizon measurement with a verifiably live channel that we have, and because the qualitative conclusion below does not rest on it alone.

With the clip and the executable guard in place, the d20 catalog arm trained for the full 4,980-step compute-optimal horizon on 8×MI355X with the channel alive on every step (commit 19–40%, abort 0.0%, codebook usage entropy > 8 bits, receipts on every step, zero ledger-less steps; 20.8 h wall, no pre-emption). The dense twin trained the same steps on the same data and silicon in 1.7 h. The twin’s per-step validation log was lost with its droplet — a custody script that tore down before verifying its upload; it now hashes every artifact and exits nonzero on any gap — so CORE on both final checkpoints is the comparison.

	catalog (live)	dense twin
final train loss	4.08	2.39
val bits-per-byte (final / min)	1.245 / 1.227	—
CORE (22 tasks, centered)	0.015	0.236
seconds / step	15.0	1.2

Table 1: Phase 5, nanochat d20, 4,980 steps; catalog run with the mis-scaled STE gradient (see text). CORE from nanochat’s `base_eval` on both final checkpoints. `results/models/d20-p5/`.

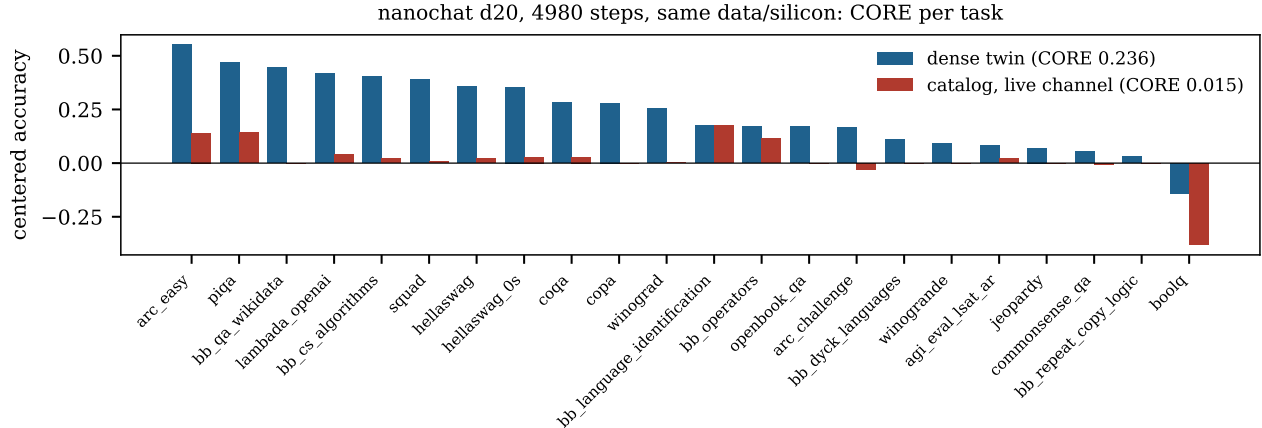


Figure 6: CORE per task. The live catalog model is at chance on every knowledge- or reasoning-bearing task (Jeopardy 0.0005 vs 0.070, Wikidata QA 0.0008 vs 0.447, LAMBADA 0.039 vs 0.418, SQuAD 0.009 vs 0.389) and matches the twin only on format-like tasks. Run with the mis-scaled STE gradient (§7).

The configuration tested is too narrow, and that conclusion does not rest on the contaminated CORE alone. At $k=8$ commits per token per write unit, each a bf16 magnitude along one of 32,768 unit vectors of dimension 20 occupying one of 64 slots, the channel carries at most 160 scalar degrees of freedom per token per layer against the twin’s 1,280 free dimensions; the training loss plateaued near 4.4 from step 1,000 to 3,000 while the twin kept descending; and the bypass scalars (Law 0) carried the trunk. We do not conclude that a transactional residual stream cannot carry a language model — we have one policy point, chosen to be verifiable rather than sufficient. The controls a capacity claim needs are missing and named: a dense bottleneck of comparable degrees of freedom, a sweep over k ($8 \rightarrow 16 \rightarrow 32 \rightarrow 64$) or over fewer, wider slots, and the catalog as an audited side-channel rather than the sole writer. Each is a rerun we have not funded.

Table 2 separates the three costs the wall-clock numbers mix: model compute, the channel’s own selection work, and verification.

8 Any-silicon records: two-phase selection

Porting to AWS Trainium2 forced a question our CUDA/HIP kernels had begged: what happens on silicon where the pinned sequential math cannot run at useful throughput? Policy 7 (*two-phase top-k*, Fig. 8) answers: the *device proposes* — bf16 matmul scores, top- M candidate features per token, XLA-compiled, never trusted — and the *reference disposes*: it recomputes the M candidate scores with the pinned math (a 1000× smaller problem) and runs the unchanged policy-6 verdict machinery. Facts remain bit-exactly

verifiable and replayable (T1/T2/apply unchanged); what is *declared* rather than verified is proposal completeness, which becomes a measured miss-rate with a preregistered threshold. **Measured on Trainium2** (trn2.3xlarge, torch_xla/Neuron): device-proposal miss-rate 0.000% across four adversarial batteries, every committed magnitude equal to the exact host recompute, and residuals equal to the reference replay bit-for-bit — the latter only after moving the apply to the host as well, because the Neuron compiler does not honor the bf16 contract (1–4 ulp drift on every touched leaf). “Reference disposes” now covers selection *and* apply. A gated bridge test proves policy 7 with $M=N_f$ is byte-identical to policy 6.

Policy 7 trains on Trainium2 at d10 (60M), catalog vs. twin. DDP over the four logical NeuronCores of a trn2.3xlarge, 400 steps at TBS 32k, the same seam, receipts and executable guard as on NVIDIA/AMD: the cataloged model reaches loss 4.667 against the dense twin’s 3.414 (same tree, same seed and data order; twin/catalog at step 100: 5.21/5.36, step 200: 4.64/5.18), with 400/400 receipts, a continuous STEP chain, 5.2×10^6 facts per step and a channel that stays alive throughout — commit 13–14% of the $k=8$ candidates per token, 86% OVERFLOW, 0% ABORT, codebook usage entropy rising from 5.0 bits at step 50 to 9.0 bits at step 350 (3,062 distinct features). Throughput is the honest number: 2.8k tok/s against the twin’s 84.5k, a $29\times$ cost, all of it in the per-write-unit host round trip (20 units $\times$ 2 microbatches per step) rather than the device proposer; a two-pass forward (propose every unit, dispose once, apply once) is the lever and is future work.

The guard fired twice before that run completed, and both firings are part of the result. The first (step

	dense twin	catalog
model FLOPs (theoretical)	$1\times$	$\approx 1.035\times$
pack / allocate / apply	—	full N_f scan on device (kernels) or top- M device + exact host dispose (policy 7)
seam: spool, T2 chain, commit	—	9.5% wall (1 GPU, 20k steps); $\sim 5\%$ at d20 $\times 8$
host round trips per step	0	0 (kernels); 40 (policy 7, 20 units $\times$ 2 microbatches)
records per step (d20)	—	3.4×10^8 ; chain + counts durable, raw sampled 1/100
wall, d20 4,980 steps, 8 $\times$ MI355X	1.7 h (1.2 s/step)	20.8 h (15.0 s/step), 12 $\times$
throughput, d10 400 steps, trn2.3xlarge	84.5k tok/s	2.8k tok/s, 29 $\times$
device memory	not measured	not measured (DBS=16 OOM at d10, DBS=4 fits)

Table 2: Where the cost is. The channel’s selection work (exact kernels) or its host round trips (policy 7) dominate; the ledger itself is under 10%. Sources: `results/models/d20-p5/`, `results/trn2-d10/`, `results/train/`.

149, entropy 6.9 bits) was a genuine collapse of our own making: the two-stage proposer described below is set-equivalent to a single top- M *except under ties*, and ties are the initial condition — nanochat zero-initialises the output projections, so every step-0 score is exactly zero and the candidate set is nothing but the tie-break. A slot-major second stage funnelled every tie into slot 0, the STE mirror then trained only slot 0, and the channel committed exactly $1/k$ from step 0 — C4 again, caught live on the third silicon. The second firing (step 149 of the corrected run, entropy 7.36 bits and rising monotonically) is the declared negative for the manifest as signed: the channel did not reach 8 bits within the 100-step warm-up plus 50-step window. We then declared a second manifest, identical except for a 200-step warm-up (threshold and window unchanged, declaration time and reason recorded in the manifest and its hash in the run’s `RUN_START`), and report the completed run under it. Both ledgers are in custody (Fig. 7).

Getting a lazy compiler to run the write unit at all took six work-arounds, none of which weakened the contract: every device op in the unit must be fixed-shape (a dense **where** mask, not a gather of the n_{commits} touched leaves, or the compiler re-traces a NEFF per step); the STE backward runs on the host from the commit facts (the Neuron PJRT backward segfaults on it); `neuronx-cc 2.26` cannot lower a top- k over a 32768-wide row for $k \geq 16$ (internal assertion; rows ≤ 16384 or $k \leq 8$ compile, **sort** is unsupported), so the proposer takes its top- M in two stages — within each slot, then over the $S \cdot M$ survivors — with an explicit composite-key tie-break (feature-ascending, like the reference; a CPU gate asserts set equality with the single top- k and zero miss-rate under exact ties); no tensor shaped $[\cdot, d_{\text{slot}}]$ may exist in the fwd+bwd graph (each is lowered to one transpose per row: 6M instructions against a 300k limit), so the apply mask is expanded to leaves on the host, the in-graph clip uses one-hot matmuls, and the codebook gradient is accumulated on the host, all-reduced over gloo and uploaded once; and the codebook stays out of the flat all-

reduce bucket. Each is a compiler limit, documented with the compile-only bisection that found it, not a change to what is verified.

The claim structure widens honestly: NVIDIA+AMD execute *bit-identical selection* (policies ≤ 6 , measured); *any* silicon — including accelerators we cannot write kernels for — executes *bit-exactly verifiable facts* under policy 7.

9 Journaled inference (instrument only)

The same seam runs at generation time: each emitted token carries the records of the writes behind it, a confidence signature computed from those records against training priors, and — above a calibrated threshold — a `LOW_CONFIDENCE` record committed *before* the token is emitted. A dead ledger degrades visibly (tokens marked uncitable), never silently. This is an instrument and a hypothesis, not a result: the planned evaluation (§11) asks whether confabulation is the inference-time signature of the structural bypass of §6. Nothing in this paper measures confabulation.

10 Related work

Discrete bottlenecks inside transformers. The closest ancestor of the catalog is Codebook Features [1]: vector-quantized bottlenecks inserted in every layer of a transformer (up to 410M parameters), giving sparse, discrete, steerable hidden states at a modest capacity cost. Backpack LMs [2] likewise change the architecture so that the interior has nameable units. Both train (or fine-tune) the bottleneck as an *interpretable* object; HYTORCH trains it as a *trans-actional* one — writes carry verdicts, the non-write (`OVERFLOW/ABORT`) is a first-class fact, and no verdict is trusted without a bit-exact reference. The channel itself is a VQ-VAE [3] trained through a straight-through estimator [4], and inherits its known pathology: codebook collapse, which the VQ literature counters with resets, normalization and better estimators

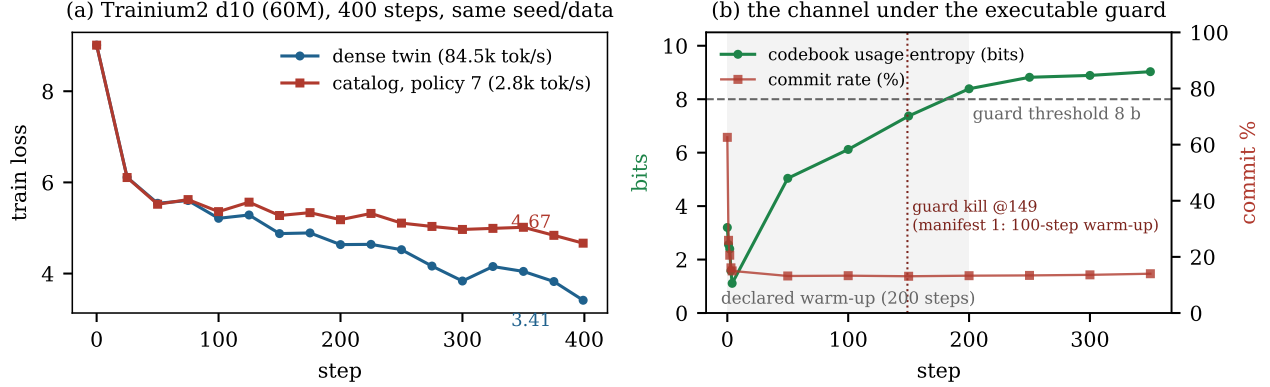


Figure 7: Third silicon. (a) Trainium2 d10, catalog under policy 7 vs. the dense twin, same seed and data: 4.67 vs. 3.41 at step 399, 2.8k vs. 84.5k tok/s. (b) The channel under the executable guard: commit rate settles at 13–14% (one commit per token per unit), codebook usage entropy climbs from 1.1 to 9.0 bits; the first manifest’s 100-step warm-up killed the run at step 149 (entropy 7.36 b, rising), the second, declared before the run, allowed 200. `results/trn2-d10/RESULT.md`.

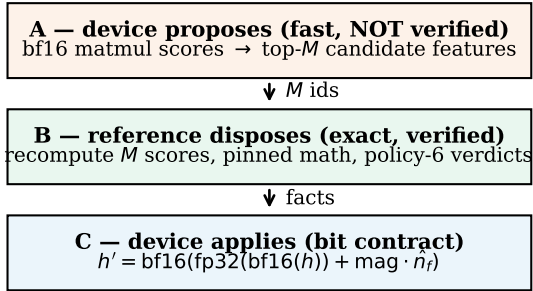


Figure 8: Policy 7: device proposes, policy disposes. The verification boundary moves; it does not weaken.

[5], and the mixture-of-experts literature with load-balancing losses [6, 7]. Our C4 finding (§6) is the same failure with a twist those settings do not have: a bypass path let the loss keep falling while the channel was dead, so the collapse was *silent*. Our answer is not a better estimator but an executable preregistered guard on commit rate and usage entropy.

Verifiable training. Proof-of-Learning [8] logs checkpoints and data indices so that a verifier can re-execute segments and land “close” to the next checkpoint; the tolerance band that absorbs GPU non-determinism is exactly what later spoofing attacks exploited [9, 10]. Zero-knowledge proofs of training [11] remove the tolerance cryptographically but at minutes of prover time per iteration on a 10M-parameter network. HYTORCH sits at a third design point: no tolerance band (the reference binary makes every verdict and residual bit-exact across vendors), no cryptographic succinctness, and a granularity neither has —

the individual 16-byte write to the residual stream, with `optimizer.step()` fail-stopped on its receipt. The receipts are tamper-evident (hash chain), not unforgeable; combining them with a PoL- or zk-style outer proof is open.

Reading the residual stream after the fact. The mechanistic interpretability line — the residual-stream framing itself [12], sparse autoencoders [13, 14, 15], transcoders and attribution graphs [16], and activation steering [17, 18] — estimates what a trained network writes by fitting a second model to its activations, then tests the estimate by intervention. Hooking frameworks such as TransformerLens [20] and the profilers and flight recorders of distributed training [21] *observe* the residual stream and the execution; none makes a write an admitted transaction or a step illegal without its record. That is the differential: HYTORCH does not observe the residual, it *admits* writes to it, and the record is the precondition of the step. The two lines are complementary, and we have one small data point on their overlap: on a wikitext-103 run the journal alone (no activations read) yields token→feature associations with lifts of 90–178 and purities up to 99.8% at 5k steps, and denying one such feature as a journalized ABORT(policy) raises the NLL of exactly the token it names (+0.0032) and nothing else (−0.0003; specificity 10.7×; `results/probe/`). The effect is small and the model young; the point is that both the reading and the intervention are records, not estimates. An SAE trained on the twin could ask what the catalog *refused* to write.

Open training and documentation requirements. Fully open training efforts [22] publish checkpoints, data and logs — transparency of artifacts,

not machine-checkable evidence of the process. The EU AI Act [23] introduces documentation obligations for general-purpose AI providers; machine-checkable training evidence of the kind built here could complement, rather than replace, such documentation. We make no claim about compliance. The preregistration discipline we borrowed [24] is standard in the empirical sciences and, to our knowledge, not previously applied to a systems paper’s own headline.

11 Limitations and future work

We name what a capacity claim needs and this paper lacks, so that none of it is inferred.

- **The corrected rerun.** Table 1 is a run with a mis-scaled STE gradient (§7). A d20 rerun with the in-graph clip, same manifest, is the experiment that would either keep CORE near 0.02 — making the capacity conclusion clean — or change it.
- **One policy point.** No sweep over k , slot width or N_f ; no catalog-as-side-channel arm; no dense bottleneck of matched degrees of freedom. “The configuration tested is too narrow” is the supported statement; “the architecture is” is not.
- **Generality of the collapse.** Shown on one host architecture whose declared bypass scalars can carry the loss (§6).
- **Throughput.** 12–29× the twin. The two-phase cost is the per-unit host round trip; a two-pass forward (propose every unit, dispose once, apply once) is the lever and is not built.
- **Retention.** Raw records are sampled (1/100 steps); chains and counts are durable. Selective retention policies are future work.
- **Integrity, not non-repudiation.** Receipts are tamper-evident hash-chain heads, forgeable by whoever runs the seam; keying or external anchoring is not built.
- **Planned evaluation: confabulation as bypass.** Signatures F1–F9 over write records, a knowledge-boundary task with mechanical labels drawn from the model’s own training shards, a preregistered AUROC ≥ 0.65 gate, and the hypothesis that confabulation is the inference-time form of §6. Designed and instrumented; not run, and it requires a cataloged model that knows something — the d20 model of §7 does not. Reported when it exists, in either direction.
- **The SFT that exposed the bug.** After the fix, SFT of the d20 catalog ran cleanly (466 steps, val bpb 0.875, channel alive); its chat evaluation is

at chance (ARC-E 25.6%, ARC-C 24.4%, MMLU 25.5%), consistent with §7. The Trainium2 d10 run (§8) is a corrected-mirror run at a smaller scale; no CORE evaluation of its checkpoints has been done.

Acknowledgements

This work trained inside nanochat [19], Andrej Karpathy’s from-scratch LLM pipeline; every d12/d20 number here was obtained with nanochat as the host trainer, the dense twin is unmodified nanochat, and CORE is nanochat’s `base_eval`. We integrate by an additive patch against a pinned commit and redistribute none of it. The Trainium2 experiments ran on the AWS Neuron team’s nanochat competition tree and toolchain; the AMD runs on DigitalOcean MI355X spot capacity.

12 Evidence table

Claim	Evidence path
Cross-vendor gate	<code>results/gates/{NVIDIA-H100,AMD-MI355X}</code>
36/36 injection	<code>results/inject/inject-summary.json</code>
1b PASS + seeds	<code>results/train/NVIDIA-H200/seeds-*</code>
LR-sweep reversal	<code>results/train/NVIDIA-H200/lrsweep-*</code>
AMD replication	<code>results/train/AMD-**-phase1b-*</code>
d12 DDP 8-rank	<code>results/nanochat/AMD-8xMI355X-d12-3.3</code>
Channel collapse	<code>results/nanochat/PHASE5-CHANNEL-*.md</code>
Kill-test resume	<code>results/nanochat/KILLTEST-PASS.md</code>
Phase-5 price (CORE)	<code>results/models/d20-p5/RESULT.md, S3</code>
Two-phase gate	<code>python/tests/test_two_phase.py</code>
Trainium2 d10 catalog vs twin	<code>results/trn2-d10/, S3</code>
Neuron compiler findings	<code>results/gates/AWS-Trainium2/</code>
Semantic probe	<code>results/probe/semantics-v0.json</code>

Table 3: Every quantitative claim maps to a committed artifact; ledger keys cited inside each NOTES file.

References

- [1] A. Tamkin, M. Taufeque, N. D. Goodman. Codebook Features: Sparse and Discrete Interpretability for Neural Networks. *ICML*, 2024.
- [2] J. Hewitt, J. Thickstun, C. D. Manning, P. Liang. Back-pack Language Models. *ACL*, 2023.
- [3] A. van den Oord, O. Vinyals, K. Kavukcuoglu. Neural Discrete Representation Learning. *NeurIPS*, 2017.
- [4] Y. Bengio, N. Léonard, A. Courville. Estimating or Propagating Gradients Through Stochastic Neurons for Conditional Computation. arXiv:1308.3432, 2013.
- [5] M. Huh, B. Cheung, P. Agrawal, P. Isola. Straightening Out the Straight-Through Estimator: Overcoming Optimization Challenges in Vector Quantized Networks. *ICML*, 2023.
- [6] N. Shazeer et al. Outrageously Large Neural Networks: The Sparsely-Gated Mixture-of-Experts Layer. *ICLR*, 2017.

- [7] W. Fedus, B. Zoph, N. Shazeer. Switch Transformers: Scaling to Trillion Parameter Models with Simple and Efficient Sparsity. *JMLR*, 23(120), 2022.
- [8] H. Jia et al. Proof-of-Learning: Definitions and Practice. *IEEE S&P*, 2021.
- [9] R. Zhang et al. “Adversarial Examples” for Proof-of-Learning. *IEEE S&P*, 2022.
- [10] C. Fang et al. Proof-of-Learning is Currently More Broken Than You Think. *IEEE EuroS&P*, 2023.
- [11] K. Abbaszadeh, C. Pappas, J. Katz, D. Papadopoulos. Zero-Knowledge Proofs of Training for Deep Neural Networks. *ACM CCS*, 2024.
- [12] N. Elhage et al. A Mathematical Framework for Transformer Circuits. *Transformer Circuits Thread*, 2021.
- [13] T. Bricken et al. Towards Monosemanticity: Decomposing Language Models With Dictionary Learning. *Transformer Circuits Thread*, 2023.
- [14] H. Cunningham, A. Ewart, L. Riggs, R. Huben, L. Sharkey. Sparse Autoencoders Find Highly Interpretable Features in Language Models. *ICLR*, 2024.
- [15] A. Templeton et al. Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet. *Transformer Circuits Thread*, 2024.
- [16] E. Ameisen et al. Circuit Tracing: Revealing Computational Graphs in Language Models. *Transformer Circuits Thread*, 2025.
- [17] A. M. Turner et al. Activation Addition: Steering Language Models Without Optimization. arXiv:2308.10248, 2023.
- [18] A. Zou et al. Representation Engineering: A Top-Down Approach to AI Transparency. arXiv:2310.01405, 2023.
- [19] A. Karpathy. nanochat: the best ChatGPT that \$100 can buy — a full-stack LLM training and inference pipeline. <https://github.com/karpathy/nanochat>, 2025.
- [20] N. Nanda, J. Bloom. TransformerLens: A Library for Mechanistic Interpretability of Generative Language Models. <https://github.com/TransformerLensOrg/TransformerLens>, 2022.
- [21] PyTorch Kineto: CPU+GPU profiling library, and the PyTorch Flight Recorder for collective operations. <https://github.com/pytorch/kineto>, 2020–2025.
- [22] D. Groeneveld et al. OLMo: Accelerating the Science of Language Models. *ACL*, 2024.
- [23] European Parliament and Council. Regulation (EU) 2024/1689 (Artificial Intelligence Act), Art. 53 and Annex XI, 2024.
- [24] B. A. Nosek, C. R. Ebersole, A. C. DeHaven, D. T. Mellor. The Preregistration Revolution. *PNAS*, 115(11), 2018.